

```
/usr/bin/python3.8 /home/student/Sreehari C/Sklearn/exp30.py
```

First few rows of the dataset:

	Id	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm	Species
0	1	5.1	3.5	1.4	0.2	Iris-setosa
1	2	4.9	3.0	1.4	0.2	Iris-setosa
2	3	4.7	3.2	1.3	0.2	Iris-setosa
3	4	4.6	3.1	1.5	0.2	Iris-setosa
4	5	5.0	3.6	1.4	0.2	Iris-setosa

Feature data (first 5 rows):

	Id	SepalLengthCm	SepalWidthCm	PetalLengthCm
0	1	5.1	3.5	1.4
1	2	4.9	3.0	1.4
2	3	4.7	3.2	1.3
3	4	4.6	3.1	1.5
4	5	5.0	3.6	1.4

Labels (first 5 rows):

```
0 Iris-setosa
1 Iris-setosa
2 Iris-setosa
3 Iris-setosa
4 Iris-setosa
```

Name: Species, dtype: object

Training features (first 5 rows):

	Id	SepalLengthCm	SepalWidthCm	PetalLengthCm
22	23	4.6	3.6	1.0
15	16	5.7	4.4	1.5
65	66	6.7	3.1	4.4
11	12	4.8	3.4	1.6
42	43	4.4	3.2	1.3

Testing features (first 5 rows):

	Id	SepalLengthCm	SepalWidthCm	PetalLengthCm
73	74	6.1	2.8	4.7
18	19	5.7	3.8	1.7
118	119	7.7	2.6	6.9
78	79	6.0	2.9	4.5
76	77	6.8	2.8	4.8

KNeighborsClassifier()

```
Array: ['Iris-versicolor' 'Iris-setosa' 'Iris-virginica' 'Iris-versicolor'
'Iris-versicolor' 'Iris-setosa' 'Iris-versicolor' 'Iris-virginica'
'Iris-versicolor' 'Iris-versicolor' 'Iris-virginica' 'Iris-setosa'
'Iris-setosa' 'Iris-setosa' 'Iris-setosa' 'Iris-versicolor'
'Iris-virginica' 'Iris-versicolor' 'Iris-versicolor' 'Iris-virginica'
'Iris-setosa' 'Iris-virginica' 'Iris-setosa' 'Iris-virginica'
'Iris-virginica' 'Iris-virginica' 'Iris-virginica' 'Iris-virginica'
'Iris-setosa' 'Iris-setosa']
```

Actual labels: 73 Iris-versicolor

```
18      Iris-setosa
118     Iris-virginica
78      Iris-versicolor
76      Iris-versicolor
31      Iris-setosa
64      Iris-versicolor
141     Iris-virginica
68      Iris-versicolor
82      Iris-versicolor
110     Iris-virginica
12      Iris-setosa
36      Iris-setosa
9       Iris-setosa
19      Iris-setosa
56      Iris-versicolor
104     Iris-virginica
69      Iris-versicolor
55      Iris-versicolor
132     Iris-virginica
29      Iris-setosa
127     Iris-virginica
26      Iris-setosa
128     Iris-virginica
131     Iris-virginica
145     Iris-virginica
108     Iris-virginica
143     Iris-virginica
45      Iris-setosa
30      Iris-setosa
Name: Species, dtype: object
```

```
Confusion Matrix: [[10  0  0]
 [ 0  9  0]
 [ 0  0 11]]
```

Accuracy: 1.0

Process finished with exit code 0